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Robust observer-based control designs for discrete nonlinear systems with disturbances

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Abstract

In this paper, we consider the robust observer-based control design problem for discrete nonlinear systems with disturbances. A new approach is proposed to provide less conservative Linear Matrix Inequality control design conditions and enhance the robustness of control designs, where the observer and controller gains are computed independently. Both the one-sided Lipschitz condition and the Linear Parameter Varying approach, which are two well-known alternatives to the classical Lipschitz condition, are considered to address nonlinearities. Two numerical examples are presented to illustrate the superiorities of the new approach over the traditional $\mathcal{H}_\infty$ approach as well as relevant works in the literature.

Keywords: Observer-based control, Robust control design, Disturbance, One-sided Lipschitz condition, Linear Parameter Varying (LPV) system, Linear matrix inequality (LMI)

1. Introduction

Feedback control is a long-established topic and has been one of the key research areas in control theory [1]. State feedback control is developed based on the assumption that full information of system states is available. This assumption, however, is hardly true in practice [2, 3]. Due to technical or economic difficulties, the full knowledge of the states can rarely be measured or available in most cases. Observer-based control, where the estimation of the true states is provided by an auxiliary system driven by the control input and measured output of the control system, has been a practical solution to this puzzle over the past decades, with many results reported in the literature [4–21].

The observer-based control problem for linear systems has been well studied based on the separation principle, where the observer and controller gains are

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computed independently [12]. On the other hand, the observer-based control problem for nonlinear counterparts is still open and challenging. In traditional approaches for nonlinear observer-based control problem, the inseparability of the observer and controller gains leads to the inevitable existence of the bilinear terms in the development of design conditions. In order to derive LMI conditions, the well-known Young inequality is usually required to linearize such bilinear terms. The Young inequality is a useful means to deal with bilinear terms. However, it further introduces conservativeness into the LMI conditions. To lessen the conservativeness produced by the Young inequality, a less conservative version of the Young inequality has been utilized instead of the classical one in some works, e.g. [15]. Nevertheless, it is interesting to question whether such bilinear terms can be avoided in the derivation of observer-based control design conditions for nonlinear systems, especially in discrete-time, i.e. can we circumvent the Young inequality in the development of nonlinear observer-based control designs and accordingly provide less conservative conditions as well as enhance the robustness of control designs? This question will be the main focus of this paper.

The well-known Lipschitz condition has been the most common approach to tackle nonlinearities in control theory. Because of its conservativeness and the inherent conservativeness in control designs based on Lipschitz condition, many approaches have been introduced as alternatives to the Lipschitz condition, e.g. the one-sided Lipschitz condition [22–30], the Linear Parameter Varying (LPV) approach [31, 32]. In the LPV approach, the Lipschitz condition is reformulated and the Lipschitz systems are converted into LPV systems. This approach is less conservative than the traditional Lipschitz approach and is useful to deal with systems with large Lipschitz constants. The class of one-sided Lipschitz systems, on the other hand, includes the class of Lipschitz systems as a special case. The one-sided Lipschitz condition also possesses some superiorities over the classical Lipschitz condition, e.g. the one-sided Lipschitz constant can be positive, negative, or zero, while the Lipschitz constant is strictly positive. It is also worth pointing out that there is a lack of works that address the observer-based control design problem for discrete one-sided Lipschitz systems.

In addition, one of the basic requirements for any good control system is the robustness against disturbances, system modeling errors, or parameter variations [1]. The presence of disturbances or errors is often unavoidable in many control systems, which can degrade the performance of control systems or even lead to instability if not taken into account. Robust control is developed to guarantee satisfactory performance of control systems subject to disturbances or errors [33–36]. To tackle this issue, $\mathcal{H}_\infty$ filtering approach has long been a practical and effective method.

Motivated by the aforementioned works, in this paper, we study the robust observer-based control problem for discrete nonlinear systems with disturbances. The main contributions of this paper can be summarized as follows:

- A new approach, which can be considered as an extension of the traditional $\mathcal{H}_\infty$ approach, is introduced with the expectations to be able to provide

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less conservative LMI conditions and enhance the robustness of control designs. In this approach, the observer and controller gains are computed independently.

- One-sided Lipschitz condition and LPV approach, which are two well-known alternatives to the restrictive Lipschitz condition, are considered to address nonlinearities.
- Using two numerical examples, we show that the new approach can indeed lead to less conservative observer-based control design conditions and improve the robustness of control systems compared to the traditional $\mathcal{H}_\infty$ approach as well as relevant works in the literature.

2. Problem Statement and Preliminaries

Consider the following class of discrete nonlinear systems with disturbances:

$$x(k+1) = Ax(k) + Bu(k) + f(x(k)) + W\omega(k), \ k \in \mathbb{N}, \tag{1a}$$

$$y(k) = Cx(k) + D\omega(k), \tag{1b}$$

where $x(k) \in \mathbb{R}^n$, $u(k) \in \mathbb{R}^m$, and $y(k) \in \mathbb{R}^p$ are the state vector, the control input vector, and the output measurement vector of system (1), respectively. $\omega(t) \in \ell_2^q$ is the exogenous disturbance vector of system (1), which can represent system modeling errors, parameter variations, or environmental noises. $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, $W \in \mathbb{R}^{n\times q}$, $C \in \mathbb{R}^{p\times n}$, and $D \in \mathbb{R}^{p\times q}$ are known, constant matrices. The nonlinear function $f(x(k))$ is assumed to satisfy

$$f(0) = 0. \tag{2}$$

Due to technical or economical difficulties, the full knowledge of the state vector $x(k)$ of system (1) may not be available [2, 3]. Observer-based controller $u(k) = -K\hat{x}(k)$ is a practical solution in this case, where K is the controller gain matrix and $\hat{x}(k) \in \mathbb{R}^n$ is the estimate of $x(k)$ computed by the observer system

$$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + f(\hat{x}(k)) + L(y(k) - C\hat{x}(k)), \tag{3}$$

where L is the observer gain matrix to be designed.
The state estimation error is defined as $e(k) = x(k) - \hat{x}(k)$. The error dynamics can be computed as

$$e(k+1) = (A - LC)e(k) + \Delta f(k) + (W - LD)\omega(k), \tag{4}$$

where $\Delta f(k) = f(x(k)) - f(\hat{x}(k))$.

Consider the observer system (3), our goal is to design the observer-based controller $u(k) = -K\hat{x}(k)$ to robustly stabilize the closed-loop system (1):

$$x(k+1) = Ax(k) - BK\hat{x}(k) + f(x(k)) + W\omega(k)$$

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$$= (A - BK)x(k) + BKe(k) + f(x(k)) + W\omega(k). \tag{5}$$

Consider the following augmented system

$$\begin{bmatrix} x(k+1) \\ e(k+1) \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x(k) \\ e(k) \end{bmatrix} + \begin{bmatrix} f(x(k)) \\ \Delta f(k) \end{bmatrix} + \begin{bmatrix} W \\ W - LD \end{bmatrix} \omega(k). \tag{6}$$

In traditional $\mathcal{H}_\infty$ approach, the gain matrices L and K are designed such that system (6) is $\mathcal{H}_\infty$ asymptotically stable, i.e. the following relation holds under zero initial condition:

$$\|x_e\|_{\ell_2^{2n}} \leq \mu \|\omega\|_{\ell_2^q}, \ \mu > 0, \tag{7}$$

where $x_e(k) = [x^T(k) \ e^T(k)]^T$, and μ is the disturbance attenuation level to be minimized.

(7) is ensured if there exists a Lyapunov-Krasovskii functional $\mathcal{V}(k) \geq 0$ such that

$$\Delta\mathcal{V}(k) + x_e^T(k)x_e(k) - \mu^2\omega^T(k)\omega(k) < 0, \tag{8}$$

where $\Delta\mathcal{V}(k) = \mathcal{V}(k+1) - \mathcal{V}(k)$.

The inseparability of the gain matrices L and K in this approach leads to the inevitable existence of the bilinear term $BK = BYP_1^{-1}$ in the derivation of observer-based control design conditions (Y, P_1 are matrix variables in design conditions). The linearization of the term BYP_1^{-1} is usually accomplished by the use of the well-known Young inequality. The Young inequality provides an convenient and effective way to linearize such bilinear terms in the development of LMI conditions. However, it clearly adds some conservativeness into the obtained conditions. A less conservative form of the Young inequality has been utilized in some recent works to reduce conservativeness, e.g. [15]. Nevertheless, it is appealing to ask: is it possible to circumvent the use of Young inequality in the development of observer-based control designs in order to provide less conservative conditions and enhance the robustness of control designs? Bear in mind that, in this paper, we introduce the following separation approach which can be considered as an extension of the traditional $\mathcal{H}_\infty$ approach.

The objective of the new approach is to design the gain matrices L and K such that the following relations simultaneously hold under zero initial conditions:

$$\|x\|_{\ell_2^n} \leq \ \mu_e\|e\|_{\ell_2^n} + \mu_{de}\|\omega\|_{\ell_2^q}, \ \mu_e > 0, \ \mu_{de} > 0, \tag{9}$$

$$\|e\|_{\ell_2^n} \leq \mu_{ds}\|\omega\|_{\ell_2^q}, \ \mu_{ds} > 0, \tag{10}$$

where μ_e, μ_{de} and μ_{ds} are attenuation levels to be optimized.
From (9) and (10), we have $\|x\|_{\ell_2^n} \leq \mu\|\omega\|_{\ell_2^q}$, where $\mu = \mu_e\mu_{ds} + \mu_{de}$, i.e. the closed-loop system (5) is $\mathcal{H}_\infty$ asymptotically stable.

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(9) and (10) are guaranteed if there exist Lyapunov-Krasovskii functionals $V_1(k) \geq 0$ and $V_2(k) \geq 0$ such that

$$\Delta V_1(k) + x^T(k)x(k) - e^T(k)P_1^{-1}e(k) - \mu_{de}^2\omega^T(k)\omega(k) < 0, \tag{11}$$

where $\mu_e = \sqrt{1/\lambda_{\min}(P_1)}$, $P_1 > 0$, and

$$\Delta V_2(k) + e^T(k)e(k) - \mu_{ds}^2\omega^T(k)\omega(k) < 0, \tag{12}$$

where $\Delta V_1(k) = V_1(k+1) - V_1(k)$, and $\Delta V_2(k) = V_2(k+1) - V_2(k)$.

Indeed, from (11) and the fact that $e^T(k)P_1^{-1}e(k) \leq \lambda_{\max}(P_1^{-1})||e(k)||^2 = 1/\lambda_{\min}(P_1)||e(k)||^2$, we have

$$\sum_{k=0}^{\infty}\Delta V_1(k) + \sum_{k=0}^{\infty}x^T(k)x(k) - \sum_{k=0}^{\infty}e^T(k)P_1^{-1}e(k) - \sum_{k=0}^{\infty}\mu_{de}^2\omega^T(k)\omega(k) \leq 0 \tag{13}$$

$$\Rightarrow V_1(\infty) - V_1(0) + ||x||_{\ell_2^n}^2 - \mu_e^2||e||_{\ell_2^n}^2 - \mu_{de}^2||\omega||_{\ell_2^q}^2 \leq 0. \tag{14}$$

Since $V_1(\infty) \geq 0$ and $V_1(0) = 0$ (zero initial condition), (14) ensures (9). Similarly, we can prove that (10) is guaranteed by (12).

In the next sections, we will utilize the new method to study the robust observer-based control design problems for one-sided Lipschitz systems and LPV systems, respectively. First, let us recall the following useful well-known results.

Lemma 1 (Young inequality)**.** *Given matrices of appropriate dimensions Ω, Φ, and $\Pi > 0$, the following inequality holds*

$$\Omega\Phi + \Phi^T\Omega^T \leq \varepsilon\Omega\Pi^{-1}\Omega^T + \varepsilon^{-1}\Phi^T\Pi\Phi, \ \forall\varepsilon \in \mathbb{R}^+. \tag{15}$$

Lemma 2 (Schur complement lemma)**.** *Given matrices of appropriate dimensions $\Omega = \Omega^T$, Φ, and $\Pi > 0$, we have $\Omega + \Phi^T\Pi\Phi < 0$ if and only if*

$$\begin{bmatrix} -\Pi^{-1} & \Phi \\ \Phi^T & \Omega \end{bmatrix} < 0. \tag{16}$$

3. One-sided Lipschitz Systems

In this section, the nonlinearity $f(x(k))$ is assumed to satisfy the one-sided Lipschitz and the quadratically inner-bounded conditions in region $\mathcal{D} \subset \mathbb{R}^n$[23]:

$$\langle f(x) - f(\hat{x}), x - \hat{x}\rangle \leq \rho||x - \hat{x}||^2, \ \forall x, \hat{x} \in \mathcal{D}, \tag{17}$$

$$||f(x) - f(\hat{x})||^2 \leq \beta||x - \hat{x}||^2 + \gamma\langle f(x) - f(\hat{x}), x - \hat{x}\rangle, \tag{18}$$

where $\rho, \beta, \gamma \in \mathbb{R}$ are known constants which can be positive, zero, or negative. ρ is known as the one-sided Lipschitz constant of the nonlinear function $f(x)$.

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Without loss of generality, we assume that $\rho \geq 0$ and $\beta \geq 0$.

Sufficient conditions in terms of LMIs, which allow us to compute the gain matrices L and K for the robust observer-based control design for one-sided Lipschitz systems, will be presented in the next theorem which utilizes the new approach. Similar to the $\mathcal{H}_\infty$ method, we wish for the attenuated levels μ_{de} and μ_{ds} to be minimal.

Theorem 1. *Given a scalar $\varepsilon_0 > 0$, the controller $u(k) = -K\hat{x}(k)$ associated with the observer system* (3) *will robustly stabilize system* (1) *by means of the approach* (9)-(10) *with a minimum attenuated level μ if there exist matrices of appropriate dimensions $P_1 > 0$, $P_2 > 0$, Y, Z, and scalars $\varepsilon_i > 0$ $(i = \overline{1,3})$, $\bar{\mu}_{de} > 0$, $\bar{\mu}_{ds} > 0$ such that the following convex optimization problems hold:* $\min(\bar{\mu}_{de})$ *subject to*

$$\begin{bmatrix} -P_1 & AP_1 - BY & BY & W & \varepsilon_1 I & 0 & 0 \\ * & -P_1 & 0 & 0 & (\gamma - \varepsilon_0)P_1 & P_1 & \sqrt{2(\varepsilon_0\rho + \beta)}P_1 \\ * & * & -P_1 & 0 & 0 & 0 & 0 \\ * & * & * & -\bar{\mu}_{de} I & 0 & 0 & 0 \\ * & * & * & * & -2\varepsilon_1 I & 0 & 0 \\ * & * & * & * & * & -I & 0 \\ * & * & * & * & * & * & -\varepsilon_1 I \end{bmatrix} < 0, \tag{19}$$

and $\min(\bar{\mu}_{ds})$ *subject to*

$$\begin{bmatrix} -P_2 & P_2A - ZC & P_2W - ZD & P_2 \\ * & \Xi & 0 & (\gamma\varepsilon_3 - \varepsilon_2)I \\ * & * & -\bar{\mu}_{ds} I & 0 \\ * & * & * & -2\varepsilon_3 I \end{bmatrix} < 0, \tag{20}$$

where $\Xi = I - P_2 + 2(\rho\varepsilon_2 + \beta\varepsilon_3)I$.
K, L and the minimized μ are obtained as $K = YP_1^{-1}$, $L = P_2^{-1}Z$ and $\mu = \sqrt{1/\lambda_{\min}(P_1)}\sqrt{\bar{\mu}_{ds}} + \sqrt{\bar{\mu}_{de}}$.

Proof. Consider the Lyapunov-Krasovskii functionals $V_1(k) = x^T(k)P_1^{-1}x(k)$, $V_2(k) = e^T(k)P_2e(k)$ $(P_1 > 0,\ P_2 > 0)$. Denote

$$A_1 = [A - BK \quad BK \quad W \quad I], \tag{21}$$
$$A_2 = [A - LC \quad W - LD \quad I], \tag{22}$$
$$\zeta_1(k) = [x^T(k) \ \ e^T(k) \ \ \omega^T(k) \ \ f^T(x(k))]^T, \tag{23}$$
$$\zeta_2(k) = [e^T(k) \ \ \omega^T(k) \ \ \Delta f^T(k)]^T. \tag{24}$$

We have

$$\begin{aligned} \Delta V_1(k) &= x^T(k+1)P_1^{-1}x(k+1) - x^T(k)P_1^{-1}x(k) \\ &= \zeta_1^T(k)A_1^TP_1^{-1}A_1\zeta_1(k) - x^T(k)P_1^{-1}x(k), \end{aligned} \tag{25}$$
$$\Delta V_2(k) = e^T(k+1)P_2e(k+1) - e^T(k)P_2e(k)$$

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$$= \zeta_2^T(k)A_2^T P_2 A_2 \zeta_2(k) - e^T(k)P_2 e(k). \tag{26}$$

From (17), (18), and (2), we obtain the following

$$0 \le 2\varepsilon_0 \varepsilon_1^{-1}(\rho x^T(k)x(k) - f^T(x(k))x(k)), \tag{27}$$
$$0 \le 2\varepsilon_1^{-1}(\beta x^T(k)x(k) + \gamma f^T(x(k))x(k) - f^T(x(k))f(x(k))), \tag{28}$$
$$0 \le 2\varepsilon_2(\rho e^T(k)e(k) - \Delta f^T(k)e(k)), \tag{29}$$
$$0 \le 2\varepsilon_3(\beta e^T(k)e(k) + \gamma \Delta f^T(k)e(k) - \Delta f^T(k)\Delta f(k)), \tag{30}$$

where ε_i $(i = \overline{0,3})$ are positive scalars. Denote

$$Y = KP_1,\ Z = P_2 L,\ \bar{\mu}_{de} = \mu_{de}^2,\ \bar{\mu}_{ds} = \mu_{ds}^2. \tag{31}$$

We have

$$\Delta V_1(k) + x^T(k)x(k) - e^T(k)P_1^{-1}e(k) - \mu_{de}^2 \omega^T(k)\omega(k) \le \zeta_1^T(k)(\Phi_1 + A_1^T P_1^{-1} A_1)\zeta_1(k), \tag{32}$$

where

$$\Phi_1 = \begin{bmatrix} \Upsilon_1 & 0 & 0 & (\gamma - \varepsilon_0)\varepsilon_1^{-1} I \\ * & -P_1^{-1} & 0 & 0 \\ * & * & -\bar{\mu}_{de} I & 0 \\ * & * & * & -2\varepsilon_1^{-1} I \end{bmatrix},$$
$$\Upsilon_1 = I - P_1^{-1} + 2(\varepsilon_0 \rho + \beta)\varepsilon_1^{-1} I.$$

By the Schur complement lemma, $\Phi_1 + A_1^T P_1^{-1} A_1 < 0$ is equivalent to

$$\begin{bmatrix} -P1 & A_1 \\ * & \Phi_1 \end{bmatrix} < 0. \tag{33}$$

Pre- and post- multiplying the left-hand side of (33) by $\mathrm{diag}\{I, P_1, P_1, I, \varepsilon_1 I\}$, i.e. applying a congruent transformation, we obtain the following equivalent condition

$$\begin{bmatrix} -P1 & AP_1 - BY & BY & W & \varepsilon_1 I \\ * & \Upsilon_2 & 0 & 0 & (\gamma - \varepsilon_0)P_1 \\ * & * & -P_1 & 0 & 0 \\ * & * & * & -\bar{\mu}_{de} I & 0 \\ * & * & * & * & -2\varepsilon_1 I \end{bmatrix} < 0, \tag{34}$$

where $\Upsilon_2 = P_1 P_1 - P_1 + 2(\varepsilon_0 \rho + \beta)\varepsilon_1^{-1} P_1 P_1$.
From (32)-(34), by Schur complement lemma, (9) is guaranteed if (19) holds. Similarly, we have

$$\Delta V_2(k) + e^T(k)e(k) - \mu_{ds}^2 \omega^T(k)\omega(k) \le \zeta_2^T(k)(\Phi_2 + A_2^T P_2 A_2)\zeta_2(k), \tag{35}$$

where

$$\Phi_2 = \begin{bmatrix} \Xi & 0 & (\gamma\varepsilon_3 - \varepsilon_2)I \\ * & -\bar{\mu}_{ds} I & 0 \\ * & * & -2\varepsilon_3 I \end{bmatrix}.$$

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By the Schur complement lemma, $\Phi_2 + A_2^T P_2 P_2^{-1} P_2 A_2 < 0$ is equivalent to

$$\begin{bmatrix} -P_2 & P_2 A_2 \\ * & \Phi_2 \end{bmatrix} < 0. \tag{36}$$

From (35)-(36), (10) is ensured if (20) holds.
From (31), K, L and μ can be obtained as $K = YP_1^{-1}$, $L = P_2^{-1}Z$ and $\mu = \sqrt{1/\lambda_{\min}(P_1)}\sqrt{\bar{\mu}_{ds}} + \sqrt{\bar{\mu}_{de}}$. □

Remark 1. As can be seen from the proof of Theorem 1, condition (11) only involves the controller gain matrix K, while condition (12) only involves the observer gain matrix L. This separation, together with the well chosen term $-e^T(k)P_1^{-1}e(k)$ in (11), allows us to manipulate the term BK into linear term without having to deal with it as a bilinear term like the tradition $\mathcal{H}_\infty$ approach, i.e. the Young inequality is not necessary in our approach.

The robust observer-based control design conditions for one-sided Lipschitz systems using the traditional $\mathcal{H}_\infty$ approach will be provided in the next theorem. The proof of this theorem is similar to the proof of Theorem 1 and will be omitted for the sake of brevity.

Theorem 2. *Given a scalar $\varepsilon_0 > 0$, the controller $u(k) = -K\hat{x}(k)$ associated with the observer system* (3) *will robustly stabilize system* (1) *by means of the traditional $\mathcal{H}_\infty$ approach* (7) *with a minimum attenuated level μ if there exist matrices of appropriate dimensions $P_1 > 0$, $P_2 > 0$, Y, Z, and scalars $\varepsilon_i > 0$ ($i = \overline{1,3}$), $\bar{\mu} > 0$ such that the following convex optimization problem holds:* $\min(\bar{\mu})$ *subject to* $\begin{bmatrix} \Gamma & \Lambda \\ \star & \Sigma \end{bmatrix} < 0$, *where*

$$\Gamma = \begin{bmatrix} -P_1 & 0 & AP_1 - BY & 0 & W & \varepsilon_1 I & 0 \\ * & -P_2 & 0 & P_2A - ZC & P_2W - ZD & 0 & P_2 \\ * & * & -P_1 & 0 & 0 & (\gamma - \varepsilon_0)P_1 & 0 \\ * & * & * & \Xi & 0 & 0 & (\gamma\varepsilon_3 - \varepsilon_2)I \\ * & * & * & * & -\bar{\mu}I & 0 & 0 \\ * & * & * & * & * & -2\varepsilon_1 I & 0 \\ * & * & * & * & * & * & -2\varepsilon_3 I \end{bmatrix},$$

$$\Xi = I - P_2 + 2(\rho\varepsilon_2 + \beta\varepsilon_3)I,$$

$$\Lambda = \begin{bmatrix} 0 & 0 & P_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{2(\varepsilon_0\rho + \beta)}P_1 & 0 & 0 & 0 & 0 \\ (BY)^T & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 & 0 \end{bmatrix}^T,$$

$$\Sigma = \mathrm{diag}\{-I, -\varepsilon_1 I, -P_1, -P_1\}.$$

K, L and the minimized μ are obtained as $K = YP_1^{-1}$, $L = P_2^{-1}Z$ and $\mu = \sqrt{\bar{\mu}}$.

Remark 2. In Theorem 1, in order to avoid bilinear term and the use of Young inequality, the scalar ε_0 is introduced as a priori rather than a variable. In

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Theorem 2, the scalar ε_0 is also introduced as a priori to circumvent such bilinear term. However, in Theorem 2, the use of Young inequality is unavoidable due to the presence of $BK = BYP_1^{-1}$. Therefore, in order to find the minimal value of μ in Theorem 1 and Theorem 2, we solve the convex optimization problems with different values of ε_0. In the numerical examples section, the following one-dimensional search strategy for ε_0 is incorporated to solve the convex optimization problems with different values of ε_0: we gradually increase the value of ε_0 from 0 to 10^4 with the step of 0.0001 on the interval $(0, 1]$, the step of 0.01 on the interval $(1, 10^2]$ and the step of 1 on the interval $(10^2, 10^4]$. Note that, ε_0 can be searched on a smaller interval with bigger steps to reduce the computational cost. The obtained minimal value of μ, however, will be less desirable. This is the trade-off choice between computational cost and system performance in practice.

4. Linear Parameter Varying Systems

In this section, the nonlinear function $f(x(k))$ is assumed to be α-Lipschitz:

$$||f(x) - f(\hat{x})|| \leq \alpha \left\| x - \hat{x} \right\|, \ \forall x, \hat{x} \in \mathbb{R}^n, \alpha > 0. \tag{37}$$

For given vectors $x = (x_1, \ldots, x_n)^T$, $y = (y_1, \ldots, y_n)^T \in \mathbb{R}^n$, let us denote the vector $x^{y_i} \in \mathbb{R}^n (i = 0, \ldots, n)$ as

$$\begin{cases} x^{y_i} = (y_1, \ldots, y_i, x_{i+1}, \ldots, x_n)^T, \ i = 1, \ldots, n, \\ x^{y_0} = x. \end{cases} \tag{38}$$

The following lemma is necessary for the development of our results in this section.

Lemma 3. *[31] Consider the nonlinear function $f = (f_1, \ldots, f_m)^T : \mathbb{R}^n \to \mathbb{R}^m$, the following properties are equivalent:*

- *Lipschitz property: f is α-Lipschitz with respect to its argument, i.e.*

$$||f(x) - f(y)|| \leq \alpha ||x - y||, \ \forall x, y \in \mathbb{R}^n. \tag{39}$$

- *Reformulated Lipschitz property: $\forall i, j = 1, \ldots, n$, there exist functions $f_{ij} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and constants $\underline{\alpha}_{ij}$ and $\bar{\alpha}_{ij}$ such that $\forall x, y \in \mathbb{R}^n$:*

$$f(x) - f(y) = \sum_{i=1}^{m} \sum_{j=1}^{n} f_{ij} H_{ij}(x - y) \tag{40}$$

and

$$\underline{\alpha}_{ij} \leq f_{ij} \leq \bar{\alpha}_{ij} \tag{41}$$

where

$$f_{ij} = f_{ij}(x^{y_{j-1}}, x^{y_j}) = \begin{cases} 0 \ \text{if} \ x_j = y_j \\ \dfrac{f_i(x^{y_{j-1}}) - f_i(x^{y_j})}{x_j - y_j} \ \text{if} \ x_j \neq y_j \end{cases}$$

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and $H_{ij}^{mn} = e_m(i)e_n^T(j)$, *where* $e_s(i) = (0, \ldots, 0, \overbrace{1}^{ith}, 0, \ldots, 0)^T \in \mathbb{R}^s$.

From (37) and (2), according to Lemma 3, there are functions $f_{ij} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, and constants $\underline{\alpha}_{ij}$, $\bar{\alpha}_{ij}$ such that

$$\underline{\alpha}_{ij} \leq f_{ij} \leq \bar{\alpha}_{ij}, \tag{42}$$

$$\Delta f(k) = f(x(k)) - f(\hat{x}(k)) = \sum_{i=1}^{n}\sum_{j=1}^{n} f_{ij}(x^{\hat{x}_{j-1}}, x^{\hat{x}_j}) H_{ij}^{nn}(x(k) - \hat{x}(k)), \tag{43}$$

$$f(x(k)) = \sum_{i=1}^{n}\sum_{j=1}^{n} f_{ij}(x^{0_{j-1}}, x^{0_j}) H_{ij}^{nn} x(k), \tag{44}$$

Denote

$$\mathcal{A}^{\diamond} = \sum_{i=1}^{n}\sum_{j=1}^{n} f_{ij}(x^{\hat{x}_{j-1}}, x^{\hat{x}_j}) H_{ij}^{nn} \in \mathbb{R}^{n\times n},$$
$$\mathcal{A}^{*} = \sum_{i=1}^{n}\sum_{j=1}^{n} f_{ij}(x^{0_{j-1}}, x^{0_j}) H_{ij}^{nn} \in \mathbb{R}^{n\times n}.$$

The estimation error dynamics (4), the closed-loop system (5), and the augmented system (6) can be rewritten as

$$e(k+1) = (A + \mathcal{A}^{\diamond} - LC)e(k) + (W - LD)\omega(k), \tag{45}$$
$$x(k+1) = (A + \mathcal{A}^{*} - BK)x(k) + BKe(k) + W\omega(k), \tag{46}$$

and

$$\begin{bmatrix} x(k+1) \\ e(k+1) \end{bmatrix} = \begin{bmatrix} A + \mathcal{A}^{*} - BK & BK \\ 0 & A + \mathcal{A}^{\diamond} - LC \end{bmatrix} \begin{bmatrix} x(k) \\ e(k) \end{bmatrix} + \begin{bmatrix} W \\ W - LD \end{bmatrix} \omega(k). \tag{47}$$

From (42), the matrices $\mathcal{A}^{\diamond}$ and $\mathcal{A}^{*}$ are in the bounded convex set $\mathcal{H}$ of which the vertices set is defined by

$$\mathcal{V}_{\mathcal{H}} = \{\Theta = (\Theta_{ij}) \in \mathbb{R}^{n\times n} : \Theta_{ij} \in \{\underline{\alpha}_{ij}, \bar{\alpha}_{ij}\}\}. \tag{48}$$

Sufficient LMIs conditions for the robust observer-based control design using the approach (9)-(10) for LPV systems will be presented in the next theorem. The theorem allows us to compute matrices K, L and the minimal μ.

Theorem 3. *The controller* $u(k) = -K\hat{x}(k)$ *associated with the observer system* (3) *will robustly stabilize system* (1) *by means of the approach* (9)-(10) *with a minimum attenuated level* μ *if there exist matrices of appropriate dimensions* $P_1 > 0$, $P_2 > 0$, Y, Z, *and scalars* $\bar{\mu}_{de} > 0$, $\bar{\mu}_{ds} > 0$ *such that the following*

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convex optimization problems hold $\forall \mathcal{A}^*, \mathcal{A}^\diamond \in \mathcal{V}_{\mathcal{H}}$*:*
$\min(\bar{\mu}_{de})$ *subject to*

$$\begin{bmatrix} -P_1 & (A+\mathcal{A}^*)P_1 - BY & BY & W & 0 \\ * & -P_1 & 0 & 0 & P_1 \\ * & * & -P_1 & 0 & 0 \\ * & * & * & -\bar{\mu}_{de}I & 0 \\ * & * & * & * & -I \end{bmatrix} < 0, \tag{49}$$

and $\min(\bar{\mu}_{ds})$ *subject to*

$$\begin{bmatrix} -P_2 & P_2(A+\mathcal{A}^\diamond) - ZC & P_2W - ZD \\ * & I - P_2 & 0 \\ * & * & -\bar{\mu}_{ds}I \end{bmatrix} < 0. \tag{50}$$

K, L and the minimized μ are obtained as $K = YP_1^{-1}$, $L = P_2^{-1}Z$ and $\mu = \sqrt{1/\lambda_{\min}(P_1)}\sqrt{\bar{\mu}_{ds}} + \sqrt{\bar{\mu}_{de}}$.

Proof. Consider the Lyapunov-Krasovskii functionals $V_1(k) = x^T(k)P_1^{-1}x(k)$, $V_2(k) = e^T(k)P_2e(k)$ ($P_1 > 0$, $P_2 > 0$). Denote

$$\bar{A}_1 = [A + \mathcal{A}^* - BK \quad BK \quad W], \tag{51}$$
$$\bar{A}_2 = [A + \mathcal{A}^\diamond - LC \quad W - LD], \tag{52}$$
$$\bar{\zeta}_1(k) = [x^T(k) \;\; e^T(k) \;\; \omega^T(k)]^T, \tag{53}$$
$$\bar{\zeta}_2(k) = [e^T(k) \;\; \omega^T(k)]^T. \tag{54}$$

We have

$$\begin{aligned} \Delta V_1(k) &= x^T(k+1)P_1^{-1}x(k+1) - x^T(k)P_1^{-1}x(k) \\ &= \bar{\zeta}_1^T(k)\bar{A}_1^TP_1^{-1}\bar{A}_1\bar{\zeta}_1(k) - x^T(k)P_1^{-1}x(k), \end{aligned} \tag{55}$$
$$\begin{aligned} \Delta V_2(k) &= e^T(k+1)P_2e(k+1) - e^T(k)P_2e(k) \\ &= \bar{\zeta}_2^T(k)\bar{A}_2^TP_2\bar{A}_2\bar{\zeta}_2(k) - e^T(k)P_2e(k). \end{aligned} \tag{56}$$

Denote

$$Y = KP_1,\ Z = P_2L,\ \bar{\mu}_{de} = \mu_{de}^2,\ \bar{\mu}_{ds} = \mu_{ds}^2. \tag{57}$$

We have

$$\Delta V_1(k) + x^T(k)x(k) - e^T(k)P_1^{-1}e(k) - \mu_{de}^2\omega^T(k)\omega(k) \le \bar{\zeta}_1^T(k)(\bar{\Phi}_1 + \bar{A}_1^TP_1^{-1}\bar{A}_1)\bar{\zeta}_1(k), \tag{58}$$

where $\bar{\Phi}_1 = \mathrm{diag}\{I - P_1^{-1}, -P_1^{-1}, -\bar{\mu}_{de}I\}$. By the Schur complement lemma, $\bar{\Phi}_1 + \bar{A}_1^TP_1^{-1}\bar{A}_1 < 0$ is equivalent to

$$\begin{bmatrix} -P1 & \bar{A}_1 \\ * & \bar{\Phi}_1 \end{bmatrix} < 0. \tag{59}$$

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Pre- and post- multiplying the left-hand side of (59) by $\mathrm{diag}\{I, P_1, P_1, I\}$, we obtain the following equivalent condition

$$\begin{bmatrix} -P1 & (A+\mathcal{A}^*)P_1 - BY & BY & W \\ * & P_1P_1 - P_1 & 0 & 0 \\ * & * & -P_1 & 0 \\ * & * & * & -\bar{\mu}_{de}I \end{bmatrix} < 0. \tag{60}$$

From (58)-(60), by Schur complement lemma, (9) is guaranteed if (49) holds. Similarly, we have

$$\Delta V_2(k) + e^T(k)e(k) - \mu_{ds}^2\omega^T(k)\omega(k) \leq \bar{\zeta}_2^T(k)(\bar{\Phi}_2 + \bar{A}_2^T P_2 \bar{A}_2)\bar{\zeta}_2(k), \tag{61}$$

where $\bar{\Phi}_2 = \mathrm{diag}\{I - P_2, -\bar{\mu}_{ds}I\}$. By the Schur complement lemma, $\bar{\Phi}_2 + \bar{A}_2^T P_2 P_2^{-1} P_2 \bar{A}_2 < 0$ is equivalent to (50), i.e. (10) is ensured if (50) holds. Note that, from (57), K, L and μ can be obtained as $K = YP_1^{-1}$, $L = P_2^{-1}Z$ and $\mu = \sqrt{1/\lambda_{\min}(P_1)}\sqrt{\bar{\mu}_{ds}} + \sqrt{\bar{\mu}_{de}}$. □

Remark 3. In [19], the authors also addressed the observer-based control design problem for LPV systems using the separation method. However, different from condition (9) in this paper which is $\|x\|_{\ell_2^n} \leq \mu_e\|e\|_{\ell_2^n} + \mu_{de}\|\omega\|_{\ell_2^q}$, the authors in [19] used the following corresponding condition

$$\|x\|_{\ell_2^n} \leq \sqrt{\nu}\|\bar{\omega}\|_{\ell_2^{m+q}}, \quad \text{where } \bar{\omega} = [(Ke)^T \ \omega^T]^T. \tag{62}$$

This condition led to the LMI condition

$$\begin{bmatrix} -P_1 & 0 & P_1(A+\mathcal{A}^*)^T - YB^T & P_1 \\ * & -\nu I & [B \quad W]^T & 0 \\ * & * & -P_1 & 0 \\ * & * & * & -I \end{bmatrix} < 0. \tag{63}$$

It is easy to see that the minimized value of ν in (63) is directly affected by not only W but also B. On the other hand, the corresponding minimized value of $\bar{\mu}_{de}$ in (49) is directly affected by W only. Numerical comparisons will be provided in Example 1 to show the difference.

The robust observer-based control design conditions for LPV systems using the traditional $\mathcal{H}_\infty$ approach will be provided in the next theorem. The proof of this theorem will be omitted for the sake of brevity as it is similar to the proof of Theorem 3.

Theorem 4. *The controller $u(k) = -K\hat{x}(k)$ associated with the observer system (3) will robustly stabilize system (1) by means of the traditional $\mathcal{H}_\infty$ approach (7) with a minimum attenuated level μ if there exist matrices of appropriate dimensions $P_1 > 0$, $P_2 > 0$, Y, Z, and scalar $\bar{\mu} > 0$ such that the following convex optimization problem holds:* $\min(\bar{\mu})$ *subject to* $\begin{bmatrix} \bar{\Gamma} & \bar{\Lambda} \\ \star & \bar{\Sigma} \end{bmatrix} < 0$, $\forall(\mathcal{A}^*, \mathcal{A}^\diamond) \in$

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$\mathcal{V}_{\mathcal{H}} \times \mathcal{V}_{\mathcal{H}}$, *where*

$$\bar{\Gamma} = \begin{bmatrix} -P_1 & 0 & (A+\mathcal{A}^*)P_1 - BY & 0 & W \\ * & -P_2 & 0 & P_2(A+\mathcal{A}^\diamond) - ZC & P_2W - ZD \\ * & * & -P_1 & 0 & 0 \\ * & * & * & I - P_2 & 0 \\ * & * & * & * & -\bar{\mu}I \end{bmatrix},$$

$$\bar{\Lambda} = \begin{bmatrix} 0 & 0 & P_1 & 0 & 0 \\ (BY)^T & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I & 0 \end{bmatrix}^T,$$

$$\bar{\Sigma} = \mathrm{diag}\{-I, -P_1, -P_1\}.$$

K, L and the minimized μ are obtained as $K = YP_1^{-1}$, $L = P_2^{-1}Z$ and $\mu = \sqrt{\bar{\mu}}$.

Remark 4. The one-sided Lipschitz condition and the LPV approach are two well-known alternatives to the restrictive Lipschitz condition in the literature. In the LPV approach, the Lipschitz condition is reformulated and the Lipschitz system is converted into LPV system where all properties of the nonlinearity are utilized. For this reason, the LPV approach can well handle nonlinear systems with large Lipschitz constants. However, the computational complexity of the LPV approach can be quite high in some cases as the number of LMI conditions in the LPV approach corresponds to the size of the vertices set. On the other hand, the class of one-sided Lipschitz systems includes the class of Lipschitz systems as a special case. The one-sided Lipschitz constant can be found positive, zero, or negative, compared to the strictly positive Lipschitz constant. While the one-sided Lipschitz condition is not as effective as the LPV approach in dealing with systems with large Lipschitz constants, it does not lead to high computational complexity in some cases as the LPV approach. Thus, depending on the the types of nonlinearities, the appropriate approach should be chosen as will be shown in Example 2.

5. Numerical Examples

In this section, we present two examples to illustrate the effectiveness and advantages of the new approach over the traditional $\mathcal{H}_\infty$ approach, as well as the edges of the new results over closely related works [12, 19] in the literature.

Example 1. Consider the single-link flexible joint robotic system [31] with disturbances in continuous-time as follows

$$\dot{x}(t) = A_c x(t) + B_c u(t) + f_c(x(t)) + W_c\omega(t),\ t \geq 0, \tag{64a}$$

$$y(t) = C_c x(t) + D_c\omega(t), \tag{64b}$$

where

$$A_c = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -48.6 & -1.25 & 48.6 & 0 \\ 0 & 0 & 0 & 1 \\ 19.5 & 0 & -19.5 & 0 \end{bmatrix},\ B_c = \begin{bmatrix} 0 \\ 21.6 \\ 0 \\ 0 \end{bmatrix},\ W_c = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

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$$C_c = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix},\ D_c = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

$$f_c(x(t)) = [0\ 0\ 0\ \ -3.33\sin(x_3(t))]^T.$$

Applying the Euler discretization method with the sampling time $t_s = 1ms = 0.001s$, we obtain the discrete-time system (1): $A = I + t_s A_c$, $B = t_s B_c$, $f(x) = t_s f_c(x)$, $W = t_s W_c$, $C = C_c$, $D = D_c$.
It is well-known that the type of nonlinearities in this example can be effectively handled by the LPV approach [31]. Hence, the results in Theorem 3 will be utilized in this example. The vertices set can be determined as

$$\mathcal{V}_{\mathcal{H}} = \left\{ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & t_s 3.33 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -t_s 3.33 & 0 \end{bmatrix} \right\}.$$

Via the Matlab LMI Control Toolbox, the convex optimization problems in Theorem 3 are feasible with the following computed gain matrices

$$K = [5705.0767\ \ 33.3035\ \ -3310.2808\ \ 401.3578],\ L = \begin{bmatrix} 0.0010 & -0.0857 \\ 0.0000 & 0.6452 \\ 0.0000 & 0.3858 \\ 0.0000 & 0.2444 \end{bmatrix},$$

and the minimized attenuation level $\mu = 1.0100$. Note that although the gain matrix K seems to be high, it actually compensates for the sampling time $t_s = 0.001s$.
For the purpose of simulation, we assume $x(0) = [0.1\ \ -0.1\ \ 0.2\ \ -0.2]^T$, $\hat{x}(0) = [0.2\ \ -0.2\ \ 0.1\ \ -0.1]^T$, and

$$\omega(k) = \begin{cases} 0.1\cos(0.5 t_s k) & \text{if} \quad k \in \bigcup\limits_{i \in \mathbb{N}\setminus\{0\}} \left(\frac{20i-3}{t_s}, \frac{20i}{t_s} \right], \\ 0 & \text{otherwise.} \end{cases}$$

The illustration of ω is shown in Figure 1. The responses of the uncontrolled system (1), i.e. $u(k) = 0$, $\forall k \in \mathbb{N}$, and the responses of the closed-loop system (5) are illustrated in Figure 2 and Figure 3, respectively. It can be observed from Figure 2 and Figure 3 that the controlled system (1) is robustly stable and the performance of the system has been noticeably improved by the observer-based controller.
Now, solving the convex optimization problem in Theorem 4 via the Matlab LMI Control Toolbox, the minimal attenuated level μ is computed as $\mu = 1.4392$. It can be seen that the result from Theorem 3 ($\mu = 1.0100$) is considerably less conservative than the result from Theorem 4 ($\mu = 1.4392$). In other words, the new approach can provide more robust observer-based control designs compared to the traditional $\mathcal{H}_\infty$ approach.
On the other hand, the LMI condition of Theorem 3.1 in [12] is infeasible for the

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system studied in this example (since we wish to robustly stabilize the closed-loop system, the matrix $\tilde{C}$ in [12] is chosen as $\tilde{C} = [I \quad I]$). As commented in Remark 3, while the minimized attenuation level in Theorem 3 is directly affected by W only, the minimized attenuation level in Theorem 3.2 [19] is directly affected by not only W but also B. To facilitate numerical comparisons between the two results, we will reconsider the system studied in this example except $B = t_s[0 \;\; b \;\; 0 \;\; 0]^T$, $b \in \mathbb{R}$. The minimized attenuation levels μ obtained by Theorem 3 and Theorem 3.2 [19] corresponding to different values of b are listed in Table 1. From Table 1, we can see that while the minimized μ obtained by Theorem 3 remains the same, the minimized μ obtained by Theorem 3.2 [19] is affected by the change of b.

Table 1: The minimized attenuation levels μ obtained by the two methods with different values of b.

b	1	10	10^2	10^3	10^4
Theorem 3	1.0100	1.0100	1.0100	1.0100	1.0100
Theorem 3.2 [19]	1.5400	1.0115	1.0048	1.0337	10.0505

Example 2. Consider the following continuous-time nonlinear system with disturbances (64) of which the nonlinearity is from the motion of a moving object [23]

$$A_c = \begin{bmatrix} -1 & 1 \\ 1.5 & -0.5 \end{bmatrix}, \; B_c = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \; W_c = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \; C_c = [0 \;\; 1], \; D = 1,$$

$$f_c(x(t)) = \begin{bmatrix} -x_1(t)(x_1^2(t) + x_2^2(t)) \\ -x_2(t)(x_1^2(t) + x_2^2(t)) \end{bmatrix}.$$

Applying the Euler discretization method with the sampling time $t_s = 1ms = 0.001s$, we get the discrete-time system (1): $A = I + t_s A_c$, $B = t_s B_c$, $f(x) = t_s f_c(x)$, $W = t_s W_c$, $C = C_c$, $D = D_c$.

In the LPV approach, the number of LMI conditions corresponds to the number of vertices. With regard to the nonlinearity in this example, the vertices set $\mathcal{V}_{\mathcal{H}_n}$ has $2^4 = 16$ vertices. Due to the large number of LMI conditions, it is not effective to use the LPV approach. Hence, the results in Theorem 1 will be utilized instead of the results in Theorem 3 in this example.

$f(x)$ is globally one-sided Lipschitz with $\rho = 0$ [23]. Consider the region $\mathcal{D} = \{x \in \mathbb{R}^2 : ||x|| \leq r\}$, $r = 10$. According to [23], $f(x)$ is quadratically inner-bounded in $\mathcal{D}$ with constants $\beta = 0$ and $\gamma = -4r^2 t_s$. Furthermore, $f(x)$ is locally Lipschitz in $\mathcal{D}$ with the Lipschitz constant $3r^2 t_s$ [23].

Using the Matlab LMI Control Toolbox, the convex optimization problems in Theorem 1 are feasible with the given positive scalar $\varepsilon_0 = 0.0372$ and the following computed gain matrices

$$K = [-5.8992 \quad -637.0308], \; L = \begin{bmatrix} 0.0020 \\ 0.0010 \end{bmatrix},$$

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and the minimized attenuation level $\mu = 1.9956$. For the purpose of simulation, we assume $x(0) = [0.1 \quad -0.1]^T$, $\hat{x}(0) = [-0.1 \quad 0.1]^T$, and $\omega(k)$ as in Example 1. The responses of the uncontrolled system (1) and the responses of the closed-loop system (5) in this example are illustrated in Figure 4 and Figure 5, respectively. It can be seen from Figure 4 and Figure 5 that the controlled system (1) is robustly stable and the performance of the system has been significantly enhanced by the observer-based controller.
Now, solving the convex optimization problem in Theorem 2 via the Matlab LMI Control Toolbox, the minimal attenuated level μ is computed as $\mu = 2.0214$. It can be seen that the result from Theorem 1 ($\mu = 1.9956$) is less conservative than the result from Theorem 2 ($\mu = 2.0214$). This means the new approach can provide more robust observer-based control designs compared to the traditional $\mathcal{H}_\infty$ approach.
On the other hand, the LMI condition of Theorem 3.1 in [12] is infeasible for the system in this example (the matrix $\tilde{C}$ in [12] is chosen as $\tilde{C} = [I \quad I]$). As mentioned earlier, the LPV approach is not suitable for the nonlinearity studied in this example. Thus, it is not suitable to apply the results in [19] to this example.

6. Conclusion

In this paper, the robust observer-based control design problem for nonlinear systems subject to disturbances in discrete-time has been studied. We have proposed a new approach, where the observer and controller gains are computed independently, to derive less conservative LMI control design conditions and improve the robustness of control designs. Both the one-sided Lipschitz condition and the LPV approach, which are two well-known alternatives to the traditional Lipschitz condition, have been considered to address nonlinearities. Via two numerical examples, the advantages of the new approach over the traditional $\mathcal{H}_\infty$ approach as well as relevant works in the literature have been illustrated.

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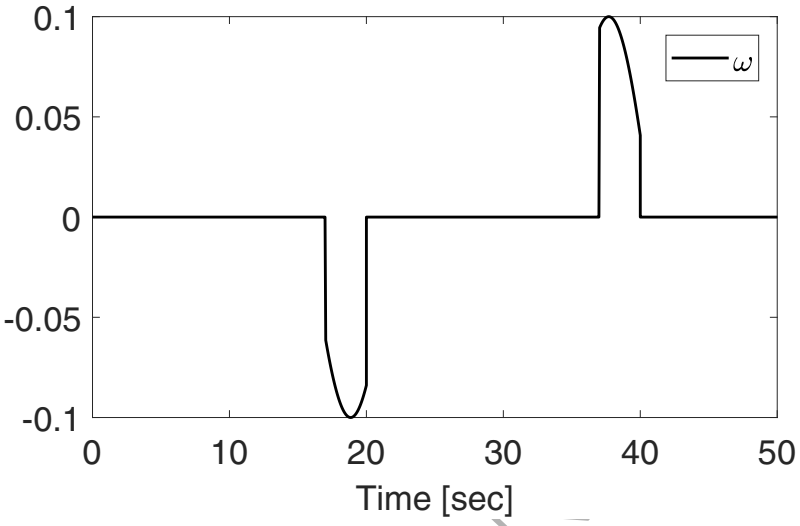

Figure 1: The disturbance ω.

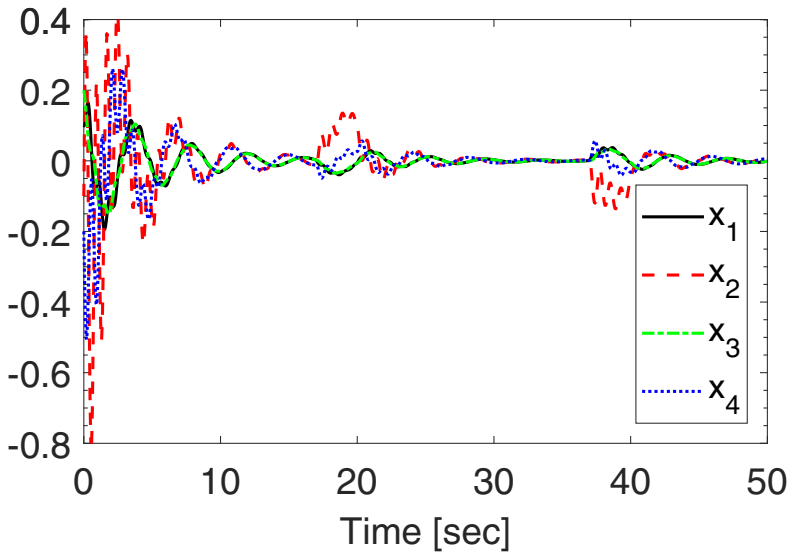

Figure 2: The responses of the uncontrolled system (1) in Example 1.

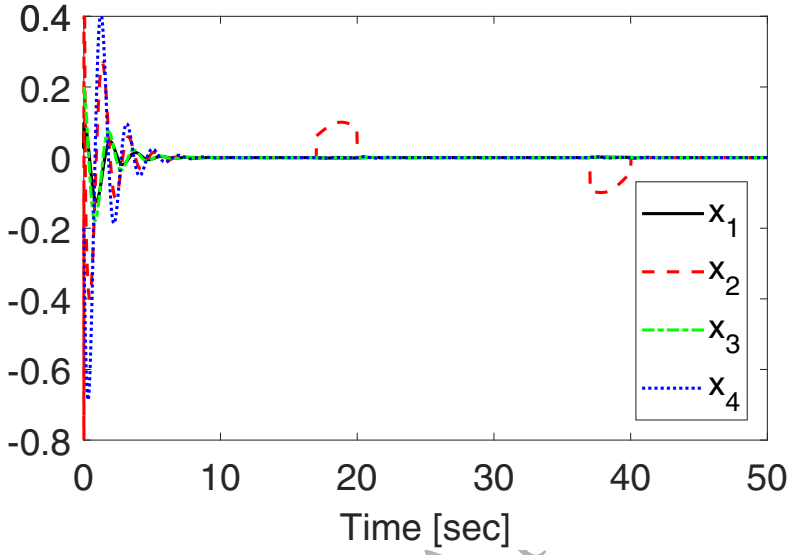

Figure 3: The responses of the controlled system (5) in Example 1.

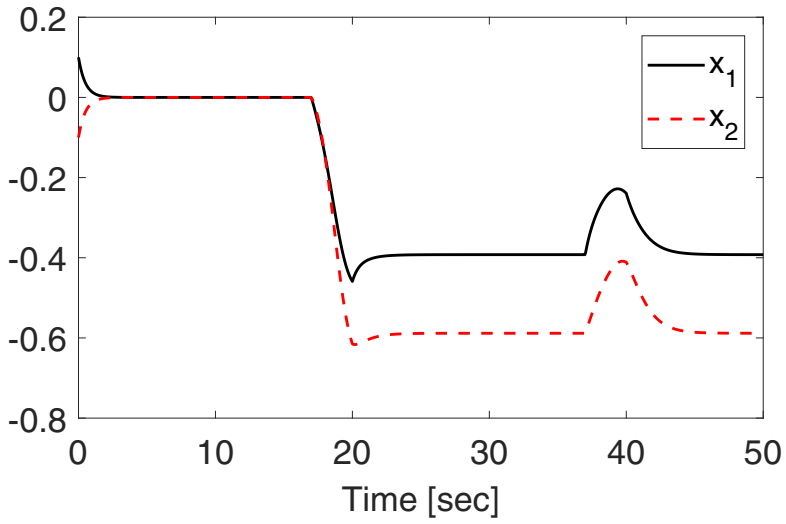

Figure 4: The responses of the uncontrolled system (1) in Example 2.

ACCEPTED MANUSCRIPT

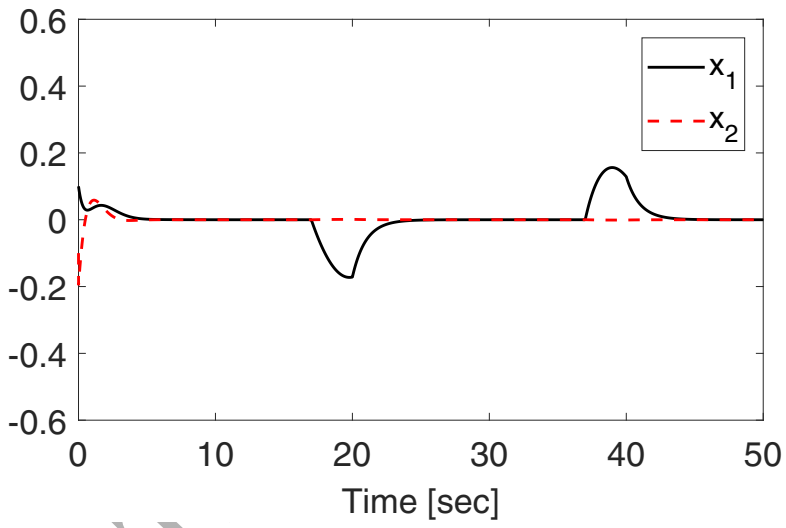

Figure 5: The responses of the controlled system (5) in Example 2.